

ChatGPT ▾

Temporärer Chat ...

TEACHING & RESEARCH DESIGN

SCENARIO 1 - SMART INSULIN PEN WITH ML

COMPONENT

THE STORY

SmartDose GmbH builds a connected insulin pen for diabetic patients.

The pen logs every injection and syncs data to a mobile app.

9 developers, running Scrum with 2-week sprints - working well for 3 years.

Mehr anzeigen ▾

TASK 1 – ANALYSE

Scrum Element	Still works?	Why / Why not?
2-week sprints	Yes, with adaptations	Short iterations are still useful for planning, integration, and risk reduction. However, ML experiments are less predictable because training results cannot be guaranteed within a sprint.
Definition of Done	Partially	Traditional criteria such as tested code and documentation still apply. Additional criteria are needed, such as data quality checks, model performance thresholds, bias evaluation, traceability, and regulatory documentation.
Backlog prioritization	Yes	Prioritization remains essential. However, the backlog must now include regulatory tasks, clinical validation activities, data acquisition, privacy work, and ML experimentation alongside software features.
Sprint Review Demo	Partially	Reviews remain valuable for showing progress. The challenge is that a model recommendation cannot simply be demonstrated as "working"; its accuracy, safety metrics, and validation evidence must also be reviewed.

Scrum Element	Still works?	Why / Why not?
Velocity tracking	Less effective	Velocity assumes similar work items can be estimated consistently. ML research, data preparation, and clinical studies involve uncertainty, making velocity a weaker predictor of future progress.

Additional new problems introduced by ML

- Safety-critical recommendations can directly harm patients.
- Outputs are probabilistic rather than deterministic.
- Patient data requires strict GDPR compliance.
- MDR approval creates significant documentation requirements.
- Models require continuous monitoring after deployment.
- The team currently lacks ML expertise.

TASK 2 – DECIDE

Situation A – Who do you hire first?

a. Which 2 do you hire first?

I would hire:

1. ML Engineer
2. Regulatory Specialist

Justification

- The ML Engineer is essential to start building the prototype and establish the technical feasibility of the dosage recommendation system.
- The Regulatory Specialist ensures that development follows MDR requirements from the beginning. Designing the system incorrectly could lead to major rework and delays during approval.

b. What risk do you consciously accept?

By delaying the other two roles:

Clinical Validator

- Risk accepted: Early prototypes may not adequately assess medical safety, potentially requiring redesign later.

Data Privacy Officer

- Risk accepted: GDPR issues may be overlooked initially, increasing the chance of delays or changes in data-handling processes.

c. When do the missing roles become critical bottlenecks?

- **Data Privacy Officer:** Before collecting, accessing, or processing patient datasets.
- **Clinical Validator:** Before any clinical evaluation, validation studies, or preparation of evidence for regulatory approval.

Situation B – Which data do you use?

a. Which dataset do you use?

I choose Hospital B.

- 5,000 patient records.
- Full informed consent for AI use.

b. Consequences for patient safety

Advantages

- Ethical and lawful use of patient data.
- Lower risk of GDPR violations.
- Greater trust from regulators and patients.

Disadvantages

- Smaller datasets may not capture rare situations.
- Increased risk that the model performs poorly on unusual patient cases.
- Additional strategies may be needed, such as collecting more consented data.

c. Consequences for the project timeline

- Development may progress more slowly.
- Additional time will likely be needed to obtain more consented data.
- Regulatory review may proceed more smoothly because data provenance is clear.

Situation C – What do you tell the CEO?

a. Which option do you choose?

Option 2: Build a shadow mode prototype.

The model operates in the background and produces recommendations that are reviewed only by physicians for research purposes. Patients never see or act upon the recommendations before approval.

Justification

- It respects MDR requirements.
- It avoids criminal violations.
- It provides valuable real-world evidence.
- Investors still see meaningful progress within six months.

b. Actual project timeline

Time	Activity
Months 0–3	Hire specialists, establish GDPR processes, acquire consented datasets, define regulatory strategy
Months 3–6	Develop prototype model and shadow mode infrastructure
Month 6	Demonstrate shadow mode system to stakeholders and clinicians
Months 6–12	Improve model performance using physician-reviewed outputs and additional data
Months 12–18	Clinical validation, preparation of MDR documentation, submission and approval activities

Time	Activity
Around Month 18	Regulatory approval completed
After approval	Patient-facing dosage recommendations may be activated

TASK 3 – REFLECT

a. Which situation was hardest to decide and why?

Situation A was the hardest because all four roles are important, yet only two hires were possible. Choosing meant deliberately accepting risks in either safety, privacy, technical capability, or regulatory compliance.

b. What would change your answer if one key assumption were different?

If the existing team already had strong regulatory experience, I would hire the **ML Engineer** and **Clinical Validator** first. In that case, ensuring medical safety early would become the higher priority.

AI ANALYSIS

Prompt used

"We are software engineering students. We designed a process for SmartDose GmbH, a company adding an ML dosage-recommendation model to a smart insulin pen. These were our decisions:

Situation A: Hire the ML Engineer and Regulatory Specialist first.

Situation B: Use Hospital B's 5,000 fully consented records.

Situation C: Build a shadow mode prototype reviewed by doctors.

Challenge our decisions. For each choice, what is the strongest argument for the option we did NOT pick? Be specific."

a. Did the AI make a valid counterargument?

Yes. A strong counterargument was that hiring a **Data Privacy Officer** immediately could prevent costly GDPR mistakes before any patient data is touched. Another valid point was that the smaller Hospital B dataset might limit the model's ability to detect rare but dangerous edge cases.

b. Would you change any of your choices?

I would keep Situations B and C unchanged because legal compliance and patient safety outweigh speed. However, I would reconsider Situation A depending on when patient data access begins, since hiring a Data Privacy Officer earlier could reduce significant compliance risks.

